

Setup Guide: Figure Eight

Time Allocation: 90 Minutes

Challenge Overview

The robot must autonomously drive in a smooth figure-eight pattern around two cones spaced one meter apart.

Environment & Arena Setup

- Select a flat, high-traction indoor surface (such as low-pile carpet or smooth gymnasium flooring) large enough to accommodate at least a 2m x 1m operating space.
- Place **two traffic cones or bright cylindrical markers** exactly **one meter apart** (measured from center-to-center of the cone bases).
- Mark a clear **starting zone** on the floor about 30 centimeters away from the cones, indicating where the robot must begin its autonomous run facing a designated direction.
- Ensure adequate, even lighting across the arena if any teams plan to use light or vision sensors for navigation.
- Set up a testing station nearby with spare tape measures, extra cones, and a designated calibration zone for teams to test their turning radii.